

Research Assignment 1

Section A: Database Fundamentals

① Main types of database

- Relational Databases (RDBMS): Store data in tables made up of rows & columns. (SQL)
- No SQL Databases: Handle large & unstructured data
- Object-Oriented Databases: Store data as objects used in programming languages
- Network Databases: Similar to hierarchical but allow multiple relationships between records
- Cloud Databases: Hosted online for scalability & easy access (Snowflake)

② RDBMS is software that manages data stored in related tables. Each table contains rows (records) & columns (fields). Relationships are created using primary & foreign keys

③ Primary Key: is a column (or group of columns) that uniquely identifies each record in a table
Foreign Key: is a column that links one table to another by referencing the primary key of a different table

④ Normalization is the process of organizing data in a database to reduce duplication & improve consistency. It divides large tables into smaller ones & establishes relationships between them

- It reduces data redundancy (no repeated information)
- Improves data accuracy & consistency
- Makes updates & more reliable

⑤ Database schema is the blueprint of the database
It defines how data is organised—tables, columns, data types

⑥ Structured — Organized in fixed rows & columns like tables. Easy to store in SQL databases,

Semi Structured — Has some structure but not fixed like tables. Uses tags

Unstructured — Has no predefined format or structure
Harder to store or analyse

⑦ Fact table stores quantitative data—numbers & measurements like sales amount, quantity sold or revenue

- A dimension table stores descriptive details such as product names, customer details

⑧ Data model is a visual representation of how data is stored, connected & processed in a system. It shows relationships between tables, fields & entities

- Helps in designing a clear data structure
- Reduces data redundancy
- Makes communication easier between developer, analysts & users

⑨ Database: Stores current, day to day operational data for quick access

Data warehouse: Stores historical & analytical data from multiple sources for reporting & analysis

Data lake: Stores raw & unstructured data for big data processing & machine learning

⑩ Data mart - is smaller, focused part of a data warehouse that serves a specific department or business - sales, marketing

Data warehouse - contains enterprise wide data

Data mart - contains only a subset of that data, relevant to one team or purpose.

SECTION B

⑪ A query language is a computer language used to communicate with & manage data in a database.

SQL is commonly used because it is standardized & universal supported by almost relational databases

- It allows both data definition & data manipulation
- It is easy to learn & powerful for managing large amounts of data

⑫ Index is special data structure that makes searching & retrieving data faster similar to an index in a book

Data without an index: The database checks every row (Slow)
The database finds the value quickly (Fast)

(13) Transaction is set of one or more database operations that are treated as a single unit of work

A - Atomicity: All operations complete successfully or none at all

C - Consistency: The database remains valid before & after a transaction

I - Isolation: Transactions don't interfere with each other

D - Durability: Once a transaction is complete changes are permanent even if the system fails.

(14) Database engine is the core software that stores, retrieves & manages data in a database. It handles reading & writing data, executing SQL commands & managing memory.

- It affects database speed & performance

(15) View - Virtual table based on a SQL query. It doesn't ^{store} data but shows data from one or more tables.

Stored procedure - A saved collection of SQL commands that can be reused

Trigger - An automatic action that runs when an event happens (like insert or delete)

(16) ETL Data is extracted then transformed (cleaned and finally loaded into the target system (Data warehouse))

ELT Data is extracted then loaded into the target system first and then transformed inside it

(17) Batch Processing — Data is collected & processed in bulk at scheduled times

Stream Processing — Data is processed continuously as it arrives in real time

Batch = periodic updates

Stream = instant updates

(18) Join type

Description

Example

INNER

Returns records that have matching values in both tables

Only customers with orders

LEFT

Returns all records from left table even if there is no match in the right

All customers, even those without orders

RIGHT

Returns all records from right table even if no match in the left

All orders even if customer info missing

FULL OUTER

Returns all records when there is a match in either table

All customers and orders matched or not

(19) Referential integrity ensures that relationships between tables remain consistent

- It prevents orphaned or invalid records

- Maintains data accuracy and reliability across tables

(20) Data redundancy - means storing the same data in multiple places

Effects

- Wastes storage space
- Cause inconsistencies
- Slow down performance because the system processes unnecessary duplicates

(21) Cloud database are hosted & managed on the internet by service providers (like AWS)
On-premise database are installed & managed locally on company-owned servers

(22) Data governance is the process of setting rules, policies & standards to ensure the data is accurate, secure & properly managed throughout its lifecycle

Importance:

- Ensure data accuracy & consistency
- Protects sensitive information
- Improve decision-making by ensuring data reliability

(23) Data integrity means that the data in a database is accurate, complete & reliable overtime

It can maintained by

- Using primary & foreign keys to maintain relationships
- Applying validation rules
- Setting constraints to prevent invalid data

(24) Data quality refers to how good, correct & useful the data is

Poor data quality leads to wrong insights & poor business decisions. In analytics bad data = data results

(25) Data analyst collects, cleans & interprets data to help businesses to make better decisions

- Extracting data from databases using SQL
- Cleaning & transforming raw data in usable formats
- Creating reports & dash boards

(26) DBA manages & maintains databases to ensure they run smoothly - secure, reliable & efficient

- Installing & configuring database software
- Monitoring database performance

(27) Data pipeline - main steps

- Extract - collect data from different sources
- Transform - Clean & format data
- Load - Store the data into a database
- Monitor → continuously track data flow & ready for analysis

(28) Performance issues when handling millions of records

- Data security risks
- High storage costs
- Data quality problems
- Backup & recovery becoming more complex

29 Database Platform	Type	Use case
MSQL	Relational	Web applications & business
PostgreSQL	Relational	Analytical & scientific
Oracle Database	Relational	Large Organizations needing high performance
Snowflake	Cloud-based	Big data analytics & cold data warehousing
MongoDB	NoSQL	Apps handling unstructured or flexible data.

30 Format	Description	Use case
CSV	Simple text file with rows & columns	Easy data exchange between tools like Excel & SQL
J-SOIN	Stores data in key value pairs human readable	Used in web APIs NoSQL databases
Parquet	Column-base binary format: very efficient for big data	Used spark, snowflake for analytics
AVRO	Compact binary format that includes data schemas	Used for data streaming & real time pipelines